Clinical-Trial Stopping Reasons and Outcomes: An Integrated Semi-Supervised and Calibrated Pipeline

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Responsibilities. Part 1 contributor: reason-label preprocessing, semi-supervised recovery, classifier experiments and predictions. Part 2 contributor: outcome modeling, calibration, cross-part analysis and integration. [TODO: Confirm full names, responsibilities and any additional team members.]

Abstract

We integrate two independently developed approaches without refitting either selected model. For five overlapping stopping reasons, a K-Means/3-neighbor hybrid followed by weighted logistic regression achieves validation sample F1 0.88675 and macro F1 0.77923. For trial success, pooled eligibility-text TF-IDF logistic regression with phase encoding achieves validation log loss 0.57375; training-only, trial-grouped sigmoid calibration modestly improves it to 0.57221. The authoritative reason predictions resolve 131 previously unavailable trial-phase rows as estimates, enabling complete recorded-reason coverage in outcome error analysis. Probability metrics improve modestly, while some small reason groups retain high false-success rates. Temporal shift, predicted-reason error and repeated validation selection limit generalization claims.

1. Methods

1.1. Data and integration controls

We use only the supplied assignment data [1]. Part 1 has 1,171 training trials (157 visible labels; 1,014 rows masked as -100), 78 validation trials and 312 test trials. Reasons are Safety, Funding, Efficacy, Recruitment and Other; several may apply. Part 2 has 8,037/1,013/1,500 train/validation/test phase rows. Validation phases I/II/III have 149/501/363 rows; test has 500 each. Repeated combined-phase trials stay within one supplied split. IDs match records but are never predictive features. No deep learning or external outcome labels are used.

The integration starts from shared main and copies only Michael's Part 2 subtree. Main's final hybrid Part 1, rather than Michael's older K=8 approach, is authoritative. Original sources and artifacts remain unchanged; newly joined reason analyses are stored separately. Neither outcome training nor calibration uses reason labels, embeddings, predictions, or their derived groups.

1.2. Part 1: recovery and classification

Standardize the 150 reason embeddings on all training rows. Fit K-Means with $K \in \{5, 10, 20\}$, k-means++ initialization, 10 starts, tolerance 10^{-4}, 300 iterations and seed 42. Compare diagonal GMMs with three starts, tolerance 10^{-3}, 200 iterations and covariance regularization 10^{-5}. For hard/soft membership r_{ik} and observed reason prior π_j, estimate

$$q_{kj} = \frac{\sum_{i \in L} r_{ik} y_{ij} + 2\pi_j}{\sum_{i \in L} r_{ik} + 2}. \quad (1)$$

Threshold the recovered scores at 0.3 or 0.5; if none crosses, assign the highest-scoring reason. Preserve all observed labels. Fit five L2 logistic regressions (liblinear, at most 1,000 iterations, seed 42) with $C \in \{0.1, 1\}$ and pseudo-row weight 0.25 or 1. Observed rows have weight 1. Final decisions use 0.5 with the same nonempty fallback.

The 70-setting initial suite includes supervised controls, 48 cluster-augmented settings, and added-year/text ablations with $C \in \{0.1, 1, 10\}$. Text/drug TF-IDF uses minimum frequency 3, up to 3,000 unigrams and sublinear term frequency; all transformations fit training only. PCA is diagnostic, not clustering input. Four Funding-only threshold controls (0.30/0.20/0.15/0.10) hold the other reasons fixed. Twenty-five local-recovery controls compare direct K-NN, K-NN recovery, and cluster/local hybrid recovery with $K \in \{3, 5, 9, 15\}$ and uniform/distance voting, plus matched supervised LR. This gives 99 evaluated configurations.

The selected hybrid equally averages scores from the best 10-cluster model and three uniform neighbors among the 157 labeled training trials. Recover masked labels at 0.3 with the fallback, then fit LR with $C = 0.1$ and pseudo-weight 0.25. Select by validation sample F1, then macro F1. K-Means/GMM and feature construction use standard scikit-learn implementations [3, 4].

1.3. Part 2: fixed model and controlled experiments

First predict each phase's training success prior. Compare 36 LR settings: three text sets (criteria; diseases/drugs/ICD; all four), $C = 0.1/1/10$, year on/off, and pooled versus independent phase models. Safely parse list-valued text; missing text becomes empty. Each field has training-fitted unigram/bigram TF-IDF with minimum frequency 3, sublinear TF, L2 normalization, at most 15,000 criteria features or 5,000 per other field. Pooled models include one-hot phase; per-phase models fit independent transforms. Year uses training-median imputation, missingness indicators and scaling. LR uses liblinear, tolerance 10^{-5}, at most 2,000 iterations and seed 5785, without class weights.

Select minimum overall validation log loss without a validation refit. The winner uses criteria plus phase, pooled $C = 1$, no year. Overall validation loss is row-weighted; report equal-phase mean separately because test phases are balanced. F1 uses probability ≥ 0.5; ROC-AUC and average precision (AP) use success probabilities. AP is not trapezoidal PR area.

1.4. Training-only probability calibration

Generate exactly one out-of-fold margin for every training row using five shuffled StratifiedGroupKFold folds, grouping by trial ID and approximately balancing phase/outcome strata (seed 5785). Each fold refits the same TF-IDF, phase encoding and $C = 1$ LR on its fit rows. Fit an unpenalized two-parameter sigmoid to these OOF margins/labels, either pooled or independently by phase. The calibrator never sees validation labels. At inference, transform margins from the original full-training base model; this is not a fold ensemble. OOF models see roughly 80% of training rows, so a margin-distribution mismatch with the full base remains possible. Sigmoid calibration follows the standard probability-calibration formulation [5].

1.5. Reason-based outcome error analysis

For outcome failures only ($y = 0$), join Part 2 trial_id to Part 1 ID. Use visible train/validation reasons first. Otherwise use the final hybrid's saved masked-train pseudo-labels or test predictions. Absent Part 1 records form a distinct no-record group; masked/prediction-unavailable rows are not negative reasons. A multi-label trial contributes to every positive reason, once within each phase. For each phase/group compute count, false-success count/fraction and mean success probability, retaining truth/prediction provenance. Compare the same group membership before/after calibration. Empty groups have undefined rates.

Part 1 method	Sample F1	Macro F1
Starter LR	0.8654	0.7067
Supervised control	0.8782	0.6990
K-Means + LR	0.8547	0.6923
GMM + LR	0.8312	0.6418
Direct K-NN	0.8632	0.6133
K-NN + LR	0.8803	0.7873
Hybrid + LR	0.8868	0.7792

Table 1. Part 1 validation results; best settings within each existing family.

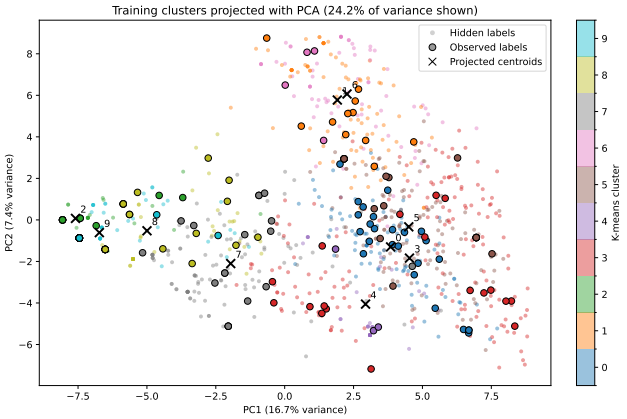


Figure 1. Existing training-only PCA diagnostic for Part 1; the first two PCs retain only 24.2% of embedding variance. Source: final Part 1 notebook.

2. Results

2.1. Part 1

The initial cluster methods underperform supervised controls (Table 1). Added year/text features do not improve initial augmented models. Funding's maximum smoothed cluster score is 0.2051, below 0.3; lowering its threshold adds 112 pseudo-positives but drops sample F1 from 0.8547 to 0.8419 through false positives. The hybrid instead recovers 16 Funding positives and identifies 2/4 Funding validation positives without false positives. It wins sample F1 0.88675; K-NN-only recovery has higher macro F1 0.78730. The recorded hybrid public Kaggle F1 is 0.87927 [2]; it is a leaderboard score, not a fully labeled test evaluation. No fitted Part 1 model file is saved, but recovered labels, validation/test predictions and source are available.

2.2. Part 2 probability and phase metrics

The phase prior has validation loss 0.67199. The best criteria model reaches 0.57375, versus 0.60076 for the best disease/drug/ICD-only setting and 0.58635 for all text. Larger C reduces training loss but worsens validation loss; the winner's training loss is 0.46903. Adding year does

Calibration	Row loss	Phase mean	Brier
Raw	0.57375	0.56816	0.19547
Pool	0.57221	0.56624	0.19493
Phase	0.57261	0.56859	0.19507

Table 2. Part 2 validation. Raw: original model; Pool/Phase: pooled or phase-specific sigmoid. Phase mean weights all phases equally.

Phase	Cal.	Loss	Brier	F1	ROC	AP
I	Raw	0.5632	0.1910	0.7717	0.7833	0.8388
I	Pool	0.5603	0.1901	0.7717	0.7833	0.8388
I	Phase	0.5706	0.1936	0.7784	0.7833	0.8388
II	Raw	0.6015	0.2073	0.6466	0.7356	0.7139
II	Pool	0.6008	0.2071	0.6463	0.7356	0.7139
II	Phase	0.6012	0.2073	0.6452	0.7356	0.7139
III	Raw	0.5399	0.1809	0.8266	0.7671	0.8607
III	Pool	0.5376	0.1802	0.8281	0.7671	0.8607
III	Phase	0.5340	0.1788	0.8235	0.7671	0.8607

Table 3. Validation metrics, F1 at 0.5. ROC denotes ROC-AUC.

not beat it. Pooled calibration improves overall loss by only 0.00154 and wins this targeted comparison (Table 2). Phase-specific calibration improves Phase III more but worsens Phase I and overall loss relative to pooled calibration (Table 3). Positive sigmoid slopes preserve within-phase rankings, so ROC-AUC/AP remain unchanged; the fixed-threshold F1 can move.

Phase III is not broadly overconfident: raw mean success probability is 66.05% versus 66.12% observed. Among 79 predictions at least 0.8, mean probability is 85.80% versus 92.41% observed success, indicating underconfidence there. The 0.4–0.5 bin overpredicts (45.21% versus 26.47%, $n = 34$), so calibration is mixed (Figure 2). Pooled Brier improves from 0.19547 to 0.19493 overall and 0.18089 to 0.18019 in Phase III. No leaderboard score is recorded for Michael's retained calibrated candidate: [TODO: add verified score and submission identity if available]. Main's separate Part 2 score 0.61046 belongs to another classifier and must not be attributed to this model.

2.3. Integrated reason coverage and errors

All 131 previously unresolved phase rows (113 unique trials) now have saved hybrid pseudo-labels; none become ground truth. Among 452 unsuccessful phase rows (398 trials), 27 use true reasons, 171 predicted reasons and 254 have no Part 1 record. Unresolved count is now zero. These estimates replace Michael's old test-reason predictions as well as fill the masked-train gap; outcome probabilities remain exactly unchanged. Phase II Recruitment improves from 18/93 false successes to 15/93, and Phase III no record from 53/82 to 52/82. Phase III Efficacy stays 7/8, all with predicted

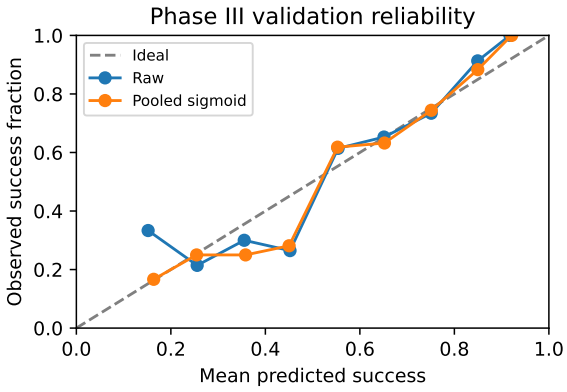


Figure 2. Phase III reliability from fixed ten-bin validation summaries. Sparse bins are noisy; the diagonal denotes perfect calibration. Source: build_tables.py.

Phase	Reason	n	T	P	Raw	Pool
I	Safety	8	1	7	0.500	0.500
I	Funding	3	1	2	0.667	0.667
I	Efficacy	4	0	4	0.750	0.750
I	Recruitment	24	2	22	0.375	0.375
I	Other	3	0	3	0.333	0.333
I	No record	28	0	0	0.464	0.464
II	Safety	11	1	10	0.273	0.273
II	Funding	4	1	3	0.000	0.000
II	Efficacy	15	0	15	0.400	0.400
II	Recruitment	93	12	81	0.194	0.161
II	Other	15	3	12	0.267	0.267
II	No record	144	0	0	0.361	0.354
III	Safety	5	1	4	0.400	0.400
III	Funding	0	0	0	–	–
III	Efficacy	8	0	8	0.875	0.875
III	Recruitment	25	5	20	0.560	0.560
III	Other	9	2	7	0.667	0.667
III	No record	82	0	0	0.646	0.634

Table 4. Integrated unsuccessful-trial groups. T/P: true/predicted reason counts; Raw/Pool: false-success fractions before/after pooled calibration. No record counts are absent records, not T/P. Groups overlap; – means undefined.

reasons; Phase I Efficacy stays 3/4. These small reason groups do not support strong rankings or causal explanations. The complete audit also retains mean predicted probabilities and the phase-specific calibration comparison.

3. Discussion

Complementary but separate tasks. Coarse cluster averages can suppress rare reasons or promote entire mixed clusters. Local neighbors provide more specific evidence; their equal blend with the unsupervised component helped

sample F1 here. Quarter-weight pseudo-labels limit but do not remove labeling errors. PCA cannot establish high-dimensional separation, and extra sparse text/drug knowledge may add noise with only 157 trusted labels. The final hybrid's 99-setting selection and only four Funding validation positives limit certainty.

Outcome calibration changes probabilities rather than vocabulary or rankings. Its small loss gain is not evidence of statistical significance. A high false-success fraction among failures alone does not establish overconfidence; calibration must consider all validation outcomes. The reason audit exposes persistent subgroup errors despite improved aggregate loss. Coverage is complete for recorded Part 1 trials, but most matched reason labels are estimates; misclassification and overlapping groups can distort comparisons. No-record trials may include completed failures, not merely trials with missing explanations.

Temporal and selection limitations. Observed train/validation start years end in 2013; test years are 2014–2021. Training has 21 missing years, validation six. Historical vocabulary, treatment mix and success rates may change. Random-by-trial validation does not measure later-year performance; the retained model has no recorded leaderboard result with which to quantify that gap. Reusing validation for 37 outcome settings and two calibrators creates selection optimism. Seven training/validation eligibility texts match exactly, despite disjoint IDs, which may also make validation easier. Combined-phase rows are correlated, and per-phase sample sizes differ. We report row and phase-macro losses distinctly. No additional training or broad search was performed during integration. A temporal holdout and repeated-seed checks are future work, not completed experiments.

References

[1] Cornell CS5785 / ORIE5750 / ECE5414 course staff. *Midterm Project*, Fall 2026. Local assignment transcription `Midterm_Project.tex`; PDF unavailable in this checkout.

[2] Team project artifacts, Fall 2026. `PART1_NOTES.md`, hybrid selection/prediction files and validation tables. Source commits and SHA-256 values: `integration/provenance.json`.

[3] Scikit-learn developers. *Clustering*, documentation. `https://scikit-learn.org/stable/modules/clustering.html`.

[4] Scikit-learn developers. *TfidfVectorizer*, API documentation. `https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html`.

[5] Scikit-learn developers. *Probability calibration*, documentation. `https://scikit-learn.org/stable/modules/calibration.html`.

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